

Digital Technologies 1.1: Create a computer program (AS92004)

Level 1 Credits 5 Assessment Internal

Achievement	Achievement with Merit	Achievement with Excellence
Create a computer program	Create a well-structured computer program.	Create a flexible and robust computer program

Student instructions

Context and issue

A factory shop is holding an end of season sale with 25% discount on all goods. To assist the workers at the checkouts a program is needed to calculate the discount for each customer.

The items have all been given the following retail prices
Jeans \$55, tshirts \$30, shoes \$45, socks \$5, skirts \$25
The factory also needs to record the total sales at each checkout.

Plan and create a program to do the following:

Start a google doc with an outline of the steps to plan your program

Make sure you identify:

Inputs, process and output

You need to use variables with 2 different data types as well as lists and constants as appropriate.

Include Iteration, Conditional statements, input and output to show your code has worked as intended

Create a flow diagram to indicate the above structures.

Ensure you document your code clearly with comments that explain what each section of code does.

- Enter the item and calculate the newprice discounting each item by 25%. Calculate the amount to pay for all the purchased items.

Data entered is the item being sold for as many items as purchased by the customer.

$\text{saleprice} = \text{retail} * 0.75$
$\text{topay} = \text{topay} + \text{saleprice}$
$\text{totalsales} = \text{totalsales} + 2\text{pay}$

- An additional discount of 10% is given if more than 3 items are purchased or if the customer spends more than \$100..

if $\text{topay} > 100$ then $\text{totaltopay} = 0.9 * \text{topay}$ if number of items > 2 then $\text{totaltopay} = 0.9 * \text{topay}$

- Output the number of items purchased, for each item: output the retail price, the discounted price and total to pay for each customer.
- Output the total sales for the day for all customers

For example, if the data entered for a customer is:

Jeans \$55 tshirt \$30	Shoes \$45, 2 x tshirt \$30,
Output would be: Number of items = 2 Jeans retail \$55, discounted price \$41.25 Tshirt retail \$30, discounted price \$ \$22.50 Total to pay = \$63.75	Output would be: Number of items = 3 Shoes retail \$45, discounted price \$33.75 Tshirt retail \$30, discounted price \$ \$22.50 Tshirt retail \$30, discounted price \$ \$22.50 Total to pay (includes extra 10% disc) = \$70.88

Total sales = \$134.63

Use the templates in Google Classroom to show your debugging processes and testing table to check for expected inputs, boundary inputs and unexpected inputs

Mark Scheme

<i>Achieved</i>	<i>Merit</i>	<i>excellence</i>
Document the program with comments	Document the program with comments that clarify the purpose of code sections	Use constants, variables, or derived values in place of literals to make the program flexible
Code works to provide a solution to the task	Code provides complete solution to the task	Code works for all inputs and is a correct solution to the task
Two or more variable data types	Independent development	checking input for validity
Variable names are meaningful	no unintended behaviour or consequences	Efficiency: no unnecessary duplication or repetition
IF structure(Condition)	Function used	Python codes of practise following such as CAPS for constants.
Loop structure (iteration) WHILE FOR	Debugging evidence and fixes	Debugging evidence of a range of examples
List structure	Testing Table shows boundary inputs tested	Testing table complete with expected, boundary and unexpected inputs
Code works for range of expected inputs		
Evidence of debugging		